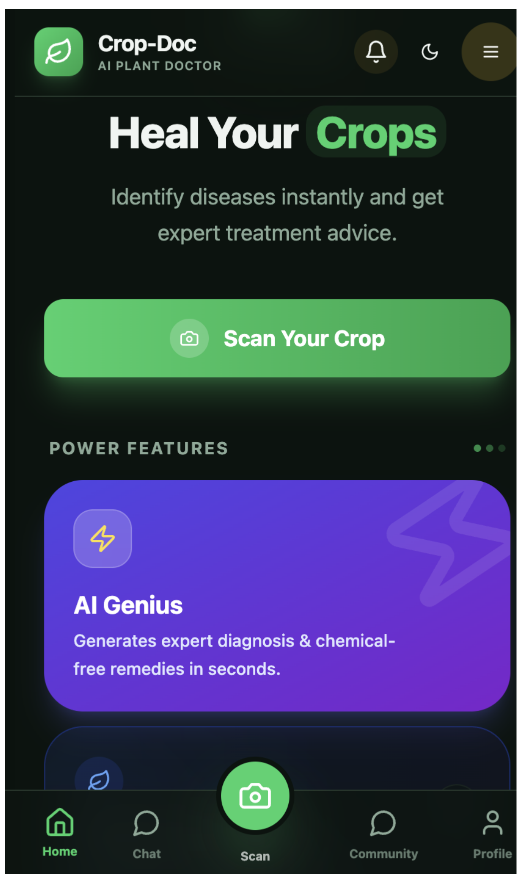
■ Crop-Doc

AI PLANT DOCTOR

“One Scan Away From a Healthier Harvest.”

Crop-Doc turns every farmer's phone into an AI plant doctor — diagnosing disease in seconds, remembering every field, warning of bad weather before it hits, and connecting a whole community of growers who help each other thrive.



■ Scan	■ Diagnose	■■ Track	■■ Protect	■ Connect

Project Documentation — Why · What · How · Tech Stack

Why Crop-Doc — The Problem

Crop-Doc exists because of three real, recurring problems farmers face every season — problems that directly cost them their crops, their time, and their peace of mind.

PROBLEM 1 — Disease detection takes too long

When a disease appears on a plant, farmers often don't notice it early — and even when they do, figuring out exactly what disease it is and what cure to use takes time they don't have. By the time the right treatment is identified, the disease has already spread, and the crop is damaged or lost entirely. In farming, speed of diagnosis is directly tied to how much of the harvest survives.

PROBLEM 2 — Tracking multiple fields is hard

Farmers often manage more than one plot of land, each growing something different. Keeping track of what was planted where — without any digital record — relies entirely on memory. There's no easy way to look back and know what was grown in a specific field, when it was planted, what it cost, or what it earned.

PROBLEM 3 — Weather catches them off guard

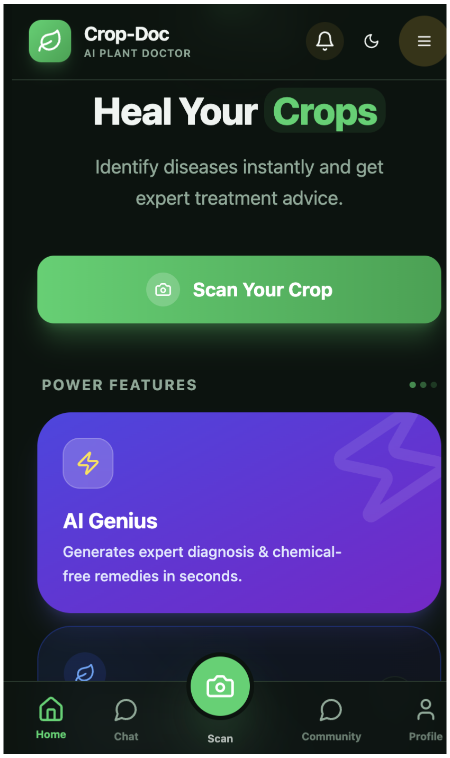
Sudden rain or bad weather can damage a harvest-ready crop or wash away a treatment that was just applied — and without any warning, farmers have no time to protect their fields. A generic regional weather forecast doesn't help if it doesn't say what's coming for their exact field, on their exact land.

The common thread: farmers don't have enough **time** or **information specific to their own fields** to protect what they've grown. Every feature in Crop-Doc is built directly around closing that gap.

Pain Point	What It Costs the Farmer
Slow disease detection	Spreading disease, damaged or lost crop, wasted treatment cost
No field record-keeping	Repeated mistakes, no proof of yield/history for loans or buyers
No hyperlocal weather warning	Crop damage or ruined treatments from unexpected rain

What Crop-Doc Is

Crop-Doc is a full-stack, mobile-friendly AI plant doctor and farm management companion. It lets a farmer scan a crop to instantly detect disease, get a cure, track every field they own, receive weather warnings before bad weather hits, and connect with other farmers — all from one app.



The Home Screen — “Heal Your Crops”

The app opens with a single clear promise: identify diseases instantly and get expert treatment advice. One tap on **Scan Your Crop** is all it takes to get started.

- **AI Genius:** generates expert diagnosis and chemical-free remedies in seconds
- A bottom navigation bar keeps Home, Chat, Scan, Community, and Profile one tap away
- Power Features carousel highlights what makes Crop-Doc different from a plain search

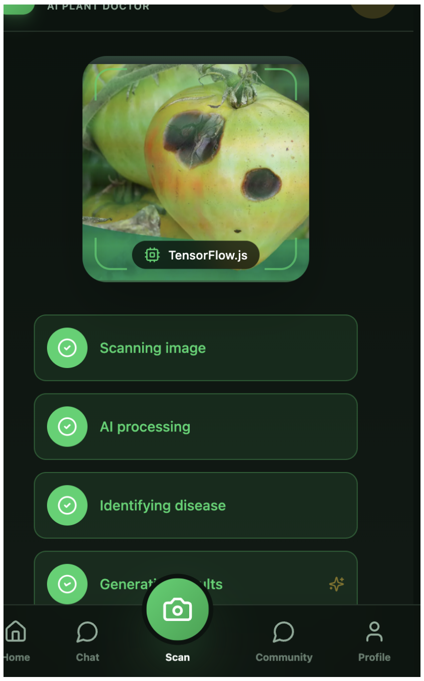
Everything the app gives a farmer

- **Instant disease detection** — scan a plant and get an AI diagnosis in seconds
- **A dedicated AI chatbot** — trained specifically to answer plant and crop-care questions
- **My Field section** — track every plot of land, what's grown there, and its exact GPS location
- **Weather alerts** — get warned about bad weather for a specific field, before it happens
- **Community forum** — post questions and connect with other farmers
- **Personal profile & activity stats** — a running record of scans, healthy crops, and posts

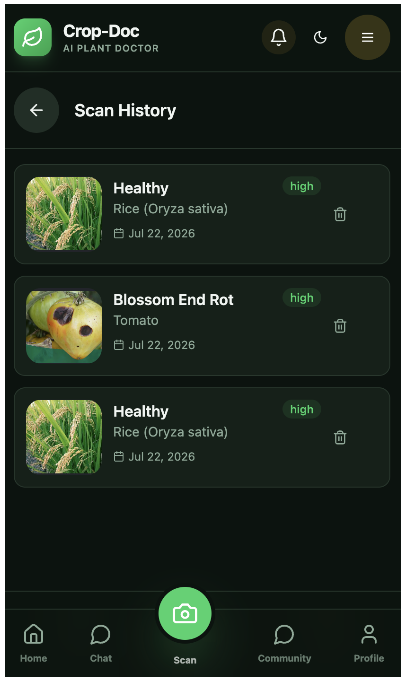
How Crop-Doc Works — Scan → Problem → Solution

The core of Crop-Doc is a simple three-step loop: a farmer notices something wrong with a plant, scans it, and gets an actionable answer — without needing to know what the disease is called or who to ask.

Step	What happens
1. Scan	The farmer opens the Scan tab and points the camera at the affected leaf or plant. TensorFlow.js runs directly in the app to ana
2. Problem	The AI processes the image and identifies the disease — the app shows live progress: Scanning image → AI processing → Ide
3. Solution	Crop-Doc generates a clear result: what the disease is, and the chemical or natural remedy to treat it — so the farmer knows e



Live scan flow: image → AI processing → disease identified.



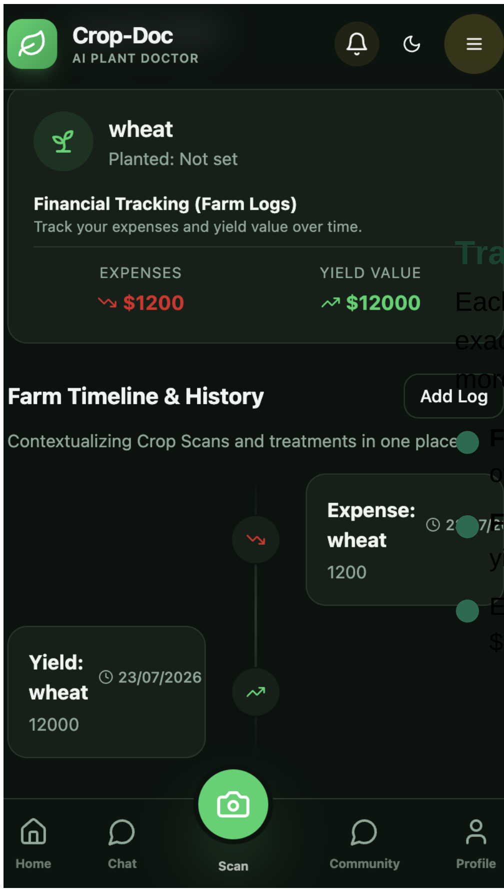
Every scan is saved automatically to Scan History with confidence level.

This entire flow directly answers Problem 1 — instead of spending hours figuring out what's wrong and searching for a cure, the diagnosis and treatment appear in seconds, right from a photo. And because every scan is saved with its result and confidence level, the farmer builds a visual, searchable history without lifting a pen.

Beyond the scan

- **History & record-keeping:** every scan is saved, giving farmers a visual, interactive record of what was scanned, when, and what the result was.
- **Dedicated AI chatbot:** for any doubt beyond a scan, farmers can ask the chatbot directly — it's trained specifically to answer plant and crop-care questions, not general topics.

My Field — Solving Problem 2



Track Every Plot Like a Ledger, Not a Memory

Each field gets its own card — crop type, planting date, and its exact location, set using GPS like a pin on Google Maps. No more guessing what was grown where.

- **Financial Tracking (Farm Logs):** log expenses and yield value over time — see profit at a glance
- **Farm Timeline & History:** every scan, treatment, expense, and yield entry lands on one chronological timeline per field
- Example shown: a wheat field logging a $1,200 expense against a $12,000 yield value

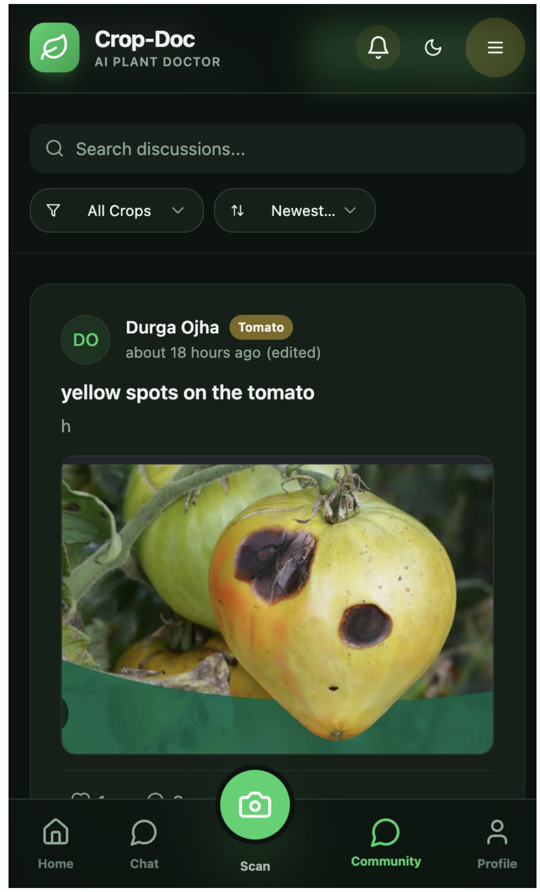
This is the direct answer to Problem 2 — instead of relying on memory to recall what's growing where, every field becomes a living record: location, cost, income, and health history, all in one place.

Community — Turning Individual Experience Into Shared Knowledge

Ask, Post, and Learn From Other Farmers

The Community tab lets farmers post a question or concern — like “yellow spots on the tomato” — with a photo attached, and get responses from others who've faced the same issue.

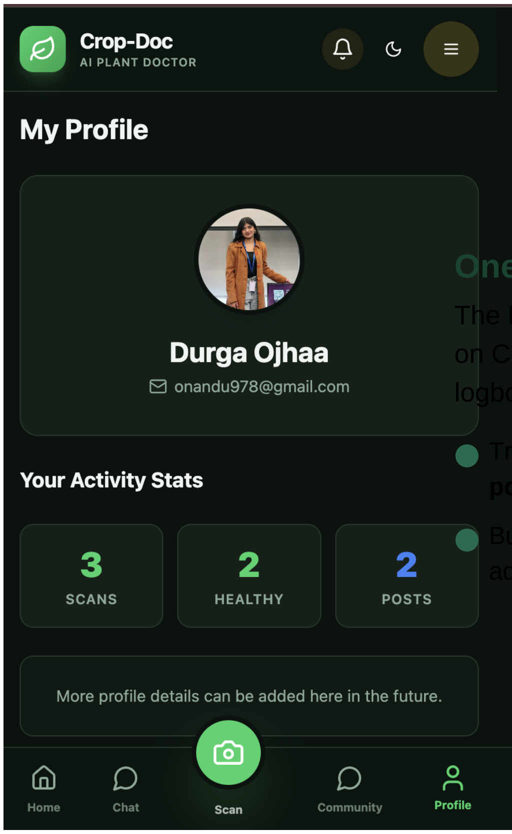
- Filter discussions by crop type or sort by newest
- Search existing discussions before posting a duplicate question
- Every post keeps the original scan photo attached for context



Weather Alerts — Solving Problem 3

If bad weather is expected for a specific field, Crop-Doc sends a notification ahead of time — for example, warning of rain risk for that exact field — so the farmer isn't caught off guard and can protect the crop or delay a treatment before it's washed away.

Profile — A Farmer's Running Record



One Place That Remembers Everything

The Profile tab keeps a running scoreboard of a farmer's activity on Crop-Doc — reinforcing trust in the app the same way a logbook builds trust in a good season.

- Tracks total **scans** performed, how many came back **healthy**, and **posts** made to the community
- Built to grow — more profile detail and achievements can be added over time

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“One Scan Away From a Healthier Harvest.”

Crop-Doc gives farmers back the one thing they never have enough of: time — time to catch a disease early, time to plan around the weather, and time saved by never having to guess where something was planted.

Tech Stack — What Powers Crop-Doc

Frontend

Technology	Role in the app
React + Vite	Core framework and build tool powering the single-page application
TypeScript	Adds strong typing across the app to catch errors early
Tailwind CSS	Utility-first styling for a fast, responsive, consistent UI
Framer Motion	Powers the smooth micro-animations seen throughout the app
React Router DOM	Handles navigation between Home, Chat, Scan, Community, and Profile
React Context API	Manages global state — authentication (AuthContext) and language (LanguageContext)
Lucide React	Icon set used consistently across the interface
TensorFlow.js	Runs the AI plant-disease detection model directly in the browser/app

Backend

Technology	Role in the app
Node.js + Express	Powers the custom REST API server (running on port 5001)
MongoDB (via Mongoose)	Stores users, scans, forum posts, field/plot data, and chat history
JWT (jsonwebtoken)	Secures user sessions and protects API routes
bcryptjs	Hashes passwords before they're stored — passwords are never stored in plain text
Multer	Handles image uploads from the Scan feature
Nodemailer + crypto	Powers secure, time-sensitive password reset emails

Core Feature-to-Tech Mapping

- **Scan & disease detection** → TensorFlow.js (on-device AI inference)
- **Dedicated AI chatbot** → a plant-specific AI model/prompted assistant, connected through the backend API
- **My Field & GPS location** → device GPS/geolocation + map-based UI, stored per field in MongoDB
- **Financial tracking (Farm Logs)** → expense/yield records stored per field in MongoDB, rendered on the timeline
- **Weather notifications** → weather API integrated on the backend, checked against each field's saved location
- **Community & posts** → MongoDB-backed forum with posts, comments, and likes, served through the API
- **Profile & activity stats** → aggregated counts of scans, healthy results, and posts per user
- **Offline scan queuing** → local caching on the frontend, synced back to the backend once online